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Effect of Six Weeks of Dura Disc and Mini-Trampoline Balance Training on Postural Sway in Athletes With Functional Ankle Instability

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Abstract. Kidgell, D.J., D.M. Horvath, B.M. Jackson, and P.J. Seymour. Effect of six weeks of dura disc and mini-trampoline balance training on postural sway in athletes with functional ankle instability. *J. Strength Cond. Res. 21(2):466–469. 2007.*—Lateral ankle sprain (LAS) is one of the most common injuries incurred during sporting activities, and effective rehabilitation programs for this condition are challenging to develop. The purpose of this research was to compare the effect of 6 weeks of balance training on either a mini-trampoline or a dura disc on postural sway and to determine if the mini-trampoline or the dura disc is more effective in improving postural sway. Twenty subjects (11 men, 9 women) with a mean age of 25.4 ± 4.2 years were randomly allocated into a control group, a dura disc training (DT) group, or a mini-trampoline (MT) group. Subjects completed 6 weeks of balance training. Postural sway was measured by subjects performing a single limb stance on a force plate. The disbursement of the center of pressure was obtained from the force plate in the medial-lateral and the anterior-posterior sway path and was subsequently used for pretest and posttest analysis. After the 6-week training intervention, there was a significant ($p < 0.05$) difference in postural sway between pre- and posttesting for both the MT (pretest = 56.8 ± 20.5 mm, posttest = 33.3 ± 8.5 mm) and DT (pretest = 41.3 ± 2.6 mm, posttest = 27.2 ± 4.8 mm) groups. There was no significant ($p > 0.05$) difference detected for improvements between the MT and DT groups. These results indicate that not only is the mini-trampoline an effective tool for improving balance after LAS, but it is equally as effective as the dura disc.

Key Words. proprioception, center of pressure, sway path, force plate

Introduction

Lateral ankle sprain (LAS) is one of the most common injuries incurred during sporting activities and is responsible for a substantial percentage of clinical referrals and emergency room visits annually (5, 7). It accounts for more time lost in participation than any other single sports-related injury and has been implicated in the development of osteoarthritis as well as other degenerative joint changes; LAS also predisposes the athlete to subsequent injuries (5). Lateral ankle sprain often results in the development of functional instability (FI), manifesting as impaired reflexes, neuromuscular firing patterns, nerve conduction velocity, and proprioception as well as losses in strength, power, and endurance (5, 9). Current research indicates that the proprioceptive loss that occurs with FI may be a major factor in the high recurrence rate of LAS, which some estimate to be as high as 80% (5).

Proprioception is such an important component of joint function because it provides an extensive amount of afferent information on the joints' internal environment, for example, tension in ligaments, intra-articular pressure, mechanical stress, and joint velocity. Without this information motor patterns that are created are not as effective and may result in the ankle being placed in an unstable situation, especially since other sources of afferent information are unable to adequately compensate for this loss.

It is for this reason that rehabilitation of the neuromuscular system is essential following LAS. Currently the most popular method involves prescribing balance activities on an unstable surface, such as an ankle disc or mini-trampoline, as these pieces of equipment are highly integrative and demanding on the neuromuscular system. A significant amount of research has examined the effectiveness of ankle disc training, demonstrating improved balance and ankle control (2, 11, 12, 17, 19); however, no such studies have ever been performed to examine the effects of undertaking balance training on a mini-trampoline, despite the fact that this method is recommended almost as frequently as the ankle disc.

Therefore, the purpose of the present study was to provide initial research on the mini-trampoline (Health Trek, Melbourne, Australia) by comparing it to a similar and well-established tool for balance training, the dura disc (AOK Health, Newcastle, Australia). Both pieces of equipment are frequently used and are recommended by clinicians because of their affordability, ease of use, transportability, and simplicity of set up. It was hypothesized that the dura disc would be more effective than the mini-trampoline at improving balance, as measured by postural sway posttraining, as the dura disc is more specific to neuromuscular retraining and because it appears to be more difficult to balance using this device.

Methods

Experimental Approach to the Problem

An experimental study design incorporating a randomized controlled design was employed. Twenty subjects (11 men, 9 women) were randomly allocated into one of the 3 following groups: dura disc training (DT; $N = 7$: 3 men and 4 women), mini-trampoline training (MT; $N = 6$: 4 men and 2 women), or a control group (C; $N = 7$: 4 men and 3 women). Those subjects assigned to the DT or MT groups completed 6 weeks of balance training. The C group completed no specific balance training; however, they were instructed to maintain their current daily activities throughout the 6-week training period.

Subjects

Twenty subjects (11 men, 9 women) who had sustained an ankle inversion injury within the last 2 years volunteered to participate in the study. Prior to participation, all subjects were fully informed of the purpose of the study, the experimental procedure, and the potential benefits and possible risks of being involved and were then asked to provide informed consent. Subjects were recruited from the university population, recreational sporting clubs in the local area, and the Victoria Police. In order to be accepted, subjects had to have suffered a LAS in the previous 2 years and had to be between 18 and 35 years of age.

Subjects that were still experiencing pain, inflammation, or limited range of movement as well as those suffering from any systemic disease (e.g., diabetes) that affects sensory-motor control were excluded from the study. Additional exclusion criteria included current involvement in an ankle rehabilitation or core stability training program. Potential subjects were also excluded if they currently had a medial ankle sprain or had never suffered an ankle sprain.

The age of the subjects ranged from 22 to 35 years, with the mean age across all subjects being 25.4 ± 4.2 years. Mean height differed slightly among the groups (169.8 ± 6.1 cm, 176.7 ± 9.7 cm, and 173.4 ± 6.0 cm, respectively) for C, MT, and DT groups. Mean weights (kg) for the groups were 69.9 ± 8.4 kg, 77.4 ± 13.9 kg, and 78.4 ± 13.8 kg for the C, MT, and DT groups, respectively. The project was approved by the University Human Research Ethics committee.

Balance Testing Procedures

Subjects were required to position themselves on top of a force plate (61 × 45 cm; Victoria University of Technologies, Victoria, Australia) with shoes off and were to maintain a unilateral stance on the injured leg for 25 seconds. The nonsupporting limb was flexed at the knee and held at approximately 90° and was not allowed to make contact with either the contralateral supporting limb or the ground during the test. Subjects were instructed to cross their arms over their chests and focus on a point 3–4 m ahead for a total of 3 trials, with an intertrial recovery period of 4 minutes. The average disbursement of the center of pressure (CoP) for the 3 trials for each subject was calculated, and the data were then used to determine the mean values for each group. This testing protocol was followed for both pre- and posttesting, with posttesting conducted no more than 5 days after the program was completed and at approximately the same time of day as the initial test, to ensure consistency. Subjects were also instructed to avoid any strenuous physical activity for at least 24 hours prior to the testing period.

Data was sampled at 75 Hz for 25 seconds, and the movement of CoP (the center of the distribution of the total force applied to the supporting surface) of the body was examined for pre- and posttest analysis. The CoP was determined as the point of application of the ground reaction force vector on the force plate surface. Center of pressure was then separated into the anterior-posterior (a-p) and medial-lateral (m-l) sway paths. The custom-written software (Victoria University of Technologies) on the force plate performed the computations; this process was automatically triggered when the subject stepped onto the top of the force plate. The components of CoP (a-p or m-l sway) were analyzed, with the most reliable parameter of CoP subsequently used for further analysis. Reliability analysis indicated that the maximum m-l sway was the most reliable measure ($R = 0.972$, $p = 0.00$). Reliability analysis was conducted to ascertain the test-retest profile of the custom-written software that calculated the parameters of CoP and center of mass. Furthermore, errors for CoP obtained via a force plate have been reported within the literature (14); therefore, it was necessary to ascertain and ensure that the measurements of CoP were in fact reliable. The mean value of the 3 trials was determined and subsequently analyzed for sway path, which is an indication of the total distance traveled by the CoP over the course of a trial and which was measured in millimeters.

Training Procedures

The training program employed a progressive overload protocol with the degree of neuromuscular demand gradually increasing over the 6-week training period. Subjects completed exercises that were specific to both the talocrural and subtalar joints; therefore, exercises included plantarflexion and dorsiflexion for the talocrural joint and controlled inversion and eversion exercises for the subtalar joint.

Both training groups completed the same closed-chain exercises; however, they were performed on different pieces of exercise equipment. The approach of using the same exercises was taken because different exercises generally require a different neural strategy and therefore produce different training responses. The aim was to investigate whether different types of rehabilitation equipment (dura disc or mini-trampoline) would cause a change in postural sway and therefore improve balance.

Subjects trained 3 times per week for 6 weeks. Exercises to be completed throughout the training period consisted of a standing static balance task, an a-p tilt (plantarflexion and dorsiflexion), and a m-l tilt (controlled inversion and eversion). All exercises were chosen for their specificity to the nature of ankle injuries. The exercises chosen were closed chain in nature (as ankle injuries occur in a closed-chain position) and were specific to training both the talocrural and subtalar joints (as these are both affected during LAS). The exercises were also specific to the testing protocol and to the disbursement of the CoP (sway path).

The training load was increased every second microcycle. Weeks 1 and 2 consisted of three 30-second static balancing exercises, three 6-repetition a-p tilt exercises, and three 6-repetition m-l tilt exercises. Weeks 3 and 4 consisted of three 60-second static balancing exercises and four each of 10-repetition a-p and m-l tilt exercises. Weeks 5 and 6 consisted of three 30-second static balancing (with eyes closed) exercises and three each of 6-repetition (with eyes closed) a-p and m-l tilt exercises.

The progressive overload approach was employed to determine the content and quality of training. The changes in training volume (sets × repetitions) were used as a process for improving muscle endurance in the musculature of the talocrural and subtalar joints, while the implementation of the practice of closing the eyes was a means of increasing the challenge of the balance tasks prescribed.

Statistical Analyses

Mean and *SD*s were calculated for the parameters obtained from the force plate, specifically, the disbursement of CoP in the m-l and a-p sway paths. Intraclass corre-

Table 1. Reliability of parameters derived from the force platform.

Sway path (mm)	Day 1 mean	Day 2 mean	Reliability	p Value
X (medial-lateral)	36.3	36.7	0.972	0.00
Y (anterior-posterior)	94.3	104.1	0.512	0.165

Table 2. Pretest and posttest data for postural sway (mean ± *SD*).

	Pretest postural sway (mm)	Posttest postural sway (mm)
Control	36.9 ± 9.9	36.7 ± 8.2
Mini-trampoline	56.8 ± 20.5	33.3 ± 8.5
Dura disc	41.3 ± 2.6	27.2 ± 4.8

lation coefficient (ICC) was used specifically to determine the intertrial reliability of the m-l (the separation of CoP in the medial and lateral path) and the a-p (the separation of CoP in the anterior and posterior path) sway path parameters derived from the CoP disbursement. The ICCs indicated that the maximum sway in the m-l direction was the most reliable measure ($R = 0.972$, $p = 0.00$), while the maximum sway in the a-p was less reliable ($R = 0.512$, $p = 0.165$); therefore, the m-l sway path was adopted as the balance measure. A 2-way analysis of variance (ANOVA) with repeated measures was used to assess the effect of training group on postural sway and to identify any differences in postural sway between the DT and MT groups. For all comparisons, a significance level of $p \leq 0.05$ was employed.

Results

Reliability of Data From the Force Platform

For the parameters derived from the force platform (disbursement of CoP in the m-l and a-p sway paths), reliability testing was conducted on 8 subjects on 2 consecutive days. Table 1 presents the ICCs for the parameters derived from the CoP disbursement. The ICCs indicated that the maximum sway in the m-l direction was the most reliable measure ($R = 0.972$, $p = 0.00$), while the maximum sway in the a-p was less reliable ($R = 0.512$, $p = 0.165$).

Force Platform Data

Table 2 displays the data derived from the force platform for each experimental group for the pre- and posttesting conditions. Data obtained from the disbursement of CoP in the m-l plane did not vary between pretest and posttest conditions for the control group. Pretest and posttest disbursement of the CoP was significantly ($p = 0.003$) different between testing conditions for the MT and DT groups.

A 2-way ANOVA with repeated measures demonstrated a significant interaction and time ($p = 0.003$) effect, indicating that there was a difference in sway path among pre- and posttesting conditions and the training group. However, there was no treatment effect, indicating that there was no significant ($p = 0.193$) difference between training group (MT or DT) and postural sway; this may be attributed to the small power observed (.233). Although these differences were not statistically significant, there did appear to be a trend evident within the data that demonstrated that there was a difference between the MT and DT groups, indicating that the mini-trampoline provided a greater improvement in static balance than the dura disc. This trend is supported by the positive percentage changes from pretest to posttest of 32 and 25.6%, respectively, for the MT and DT groups.

Discussion

The main findings in this investigation are (a) that 6 weeks of a balance training program improved postural sway among persons with ankle instability and (b) that balance training on a mini-trampoline is just as effective as balance training on a dura disc (ankle disc). Research has demonstrated reduced postural sway and proprioceptive deficits among persons with ankle instability (1, 10, 16), and evidence indicates that this proprioceptive deficit can be restored via a balance training program (10, 17, 19).

The finding for the dura disc supports the findings associated with current research (10, 12, 16, 17); however, the significant change in postural sway within the MT group is of particular importance, as no other study has demonstrated such changes. In fact, this finding supports previously unconfirmed recommendations of using the mini-trampoline as a tool for rehabilitation after sustaining LAS.

A possible explanation for the effectiveness of the training program is the focus on prescribing exercises that are endurance based rather than strength based. Research has demonstrated that endurance in surrounding musculature is an important component of balance ability; however, strength training is not (6). Many studies investigating a link between impaired balance ability and strength deficits have determined that proprioceptive deficits occur independently of muscle weakness (7). Furthermore, the findings of Powers et al. (13) support this notion. They found that after 6 weeks of strength training the muscles of the talocrural joint, there were no changes in postural sway.

In the current study, it could be suggested that the training improvements obtained in postural sway may be attributed to the specificity of the exercise prescribed. Both the DT and MT groups completed exactly the same exercises. The exercises had an endurance focus, which is an important component for balance ability (16); the exercises also trained the muscles of the subtalar and talocrural joint and were all closed chain in nature. Poor muscular endurance within the muscles that evert and invert the foot has been associated with ankle instability (8), as has poor endurance of the muscles that dorsiflex and plantarflex the ankle. It is important to exercise the muscles of these joints, as a LAS often involves other joints and ligaments within the ankle complex, in particular the subtalar joint (10). Since the exercises were performed on pieces of equipment that challenge the neuromuscular system and comprised specific movement patterns with regard to both the talocrural and subtalar musculature, it is likely that muscle endurance and sensorimotor control improved, even though these measurements were not obtained. Also, it is possible that the 6-week training program may have improved reaction times within the muscles of the ankle complex. Research has demonstrated delayed reaction times in the tibialis anterior and posterior after ankle disc training (9, 12, 17). While the present investigation did not measure reaction times, the practical implication for delayed activity in these muscles would be enhanced efficiency in controlling excessive inversion, which would therefore improve joint stability (9).

The results of our study contradict the results of the study of Powers et al. (13), who found no effect of 6 weeks of balance training on improved postural stability characteristics among subjects with ankle instability. It is possible that no significant effects were found because of the exercises prescribed.

This discrepancy may be attributed to the open-chained T-band kick exercises prescribed, which focused solely on plantarflexion and dorsiflexion of the talocrural joint, ignoring the movements of inversion and eversion at the subtalar joint, which is also known to be damaged during LAS. A major limitation of open-chain rehabilitation, in general or specific to the ankle joint, is the artificial nature of the T-band kick exercises. While these exercises may improve the ankle deficit, the principle of specific adaptation to imposed demands indicates that transfer of these improved neuro-physiological capabilities to complete the activities of daily living and sport is less than optimal (4). Furthermore, this discrepancy may also be due to the fact that the testing procedures were conducted in a closed-chain position or due to the lack of incremental volume throughout the 6-week training period.

In the present investigation, the exercises prescribed specifically trained the muscles of the ankle in a closed-chain position, therefore allowing for the integration of the hip and ankle musculature during a single leg stance. Weakness in the gluteus medius is associated with a change in the center of gravity when standing on a single leg (3).

There is some evidence to indicate that persons with ankle instability utilize a different postural control strategy when compared to healthy controls (2, 15). Tropp and Odenrick (18) noted that subjects with ankle instability had a greater CoP excursion and displayed a greater tendency to utilize the hip joint when correcting postural control. It is possible that the changes in postural sway seen with the training groups may have been due to improved pelvic stabilization, therefore reducing the likelihood that the hip joint would be utilized during postural control correction. Furthermore, all of the exercises prescribed utilized a standing single leg position combined with specific movements of the ankle complex, making the training program specific to the testing procedures.

Interestingly, the 2-way repeated-measures ANOVA showed no significant difference ($p = 0.165$) between the effectiveness of the dura disc and the mini-trampoline, which does not support our hypothesis that the dura disc would be more effective. This may indicate that the mini-trampoline provides just as much demand on the neuromuscular system as the dura disc and that either device can be recommended to produce the same degree of improved balance when measured by postural sway. Based on these results, either device could be recommended for LAS rehabilitation, so decisions may be based on other factors, such as price and the space available for the equipment to be utilized.

However, it must be highlighted that the present investigation only provides preliminary data with regard to the effectiveness of the mini-trampoline for postural sway. The observed power of the study was small (0.233) because of the sample size that had to be chosen, since equipment availability was limited and time constraints restricted the duration of the training intervention to 6 weeks.

Practical Applications

Strength and conditioning professionals play an important role in the rehabilitation process of athletes. This study has demonstrated that 6 weeks of balance training on either a dura disc or a mini-trampoline is effective in improving postural control among persons with ankle instability. Furthermore, the change in postural sway seen within the mini-trampoline group is an important finding, since to our knowledge this is the first study to have exclusively examined the effects of balance training on the mini-trampoline and its effect on postural sway. This may indicate that the mini-trampoline provides just as much demand on the neuromuscular system as the dura disc and that either device can be recommended to produce the same level of improved balance when measured by postural sway.

References

1. Bernier, J.N., and D.H. Perrin. Effect of coordination training on proprioception of the functionally unstable ankle. *J. Orthop. Sports Phys. Ther.* 27:264–275. 1998.
2. Gauffin, H., H. Tropp, and P. Odenrick. Effect of ankle disc training on postural control in patients with functional instability of the ankle joint. *Int. J. Sports Med.* 9:141–144. 1988.
3. Gribble, P.A., J. Hertel, C.R. Denegar, and W.E. Buckley. Effect of hip and ankle muscle fatigue on unipedal postural control. *J. Electromyogr. Kinesiol.* 14:641–646. 2004.
4. Harter, R.A. Clinical rationale for closed kinetic chain activities in functional testing and rehabilitation of ankle pathologies. *J. Sports Rehabil.* 5:13–24. 1996.
5. Hertel, J. Functional anatomy, pathomechanics, and pathophysiology of lateral ankle instability. *J. Athletic Train.* 37:364–375. 2002.
6. Kaminski, T.W., B.D. Buckley, M.E. Powers, T.J. Hubbard, and C. Ortiz. Effect of strength and proprioception training on eversion to inversion strength ratios in subjects with unilateral functional ankle instability. *Br. J. Sports Med.* 37:410–416. 2004.
7. Kaminski, T.W., and H.D. Hartsell. Factors contributing to chronic ankle instability: A strength perspective. *J. Athletic Train.* 37:394–405. 2002.
8. Kaumeyer, G., and T. Malone. Ankle injuries: Anatomical and biomechanical considerations necessary for the development of an injury prevention program. *J. Orthop. Sports Phys. Ther.* 1:171–177. 1980.
9. Konradsen, L. Sensori-motor control of the uninjured and injured human ankle. *J. Electromyogr. Kinesiol.* 12:199–203. 2002.
10. Mann, G., M. Nyska, N. Constantini, Y. Matan, P. Renström, and S.A. Lynch. Mechanics of injury, clinical presentation, and staging. In: *The Unstable Ankle.* M. Nyska and G. Mann, eds. Champaign, IL: Human Kinetics, 2002. pp. 54–61.
11. Matsusaka, N., S. Yokoyama, T. Tsurusaki, S. Inokuchi, and M. Okita. Effect of ankle disc training combined with tactile stimulation to the leg and foot on functional instability of the ankle. *Am. J. Sports Med.* 29: 25–30. 2001.
12. Osborne, M.D., L. Chou, E.R. Laskowski, J. Smith, and K.R. Kaufman. The effect of ankle disk training on muscle reaction time in subjects with a history of ankle sprain. *Am. J. Sports Med.* 29:627–632. 2001.
13. Powers, M.E., B.D. Buckley, T.W. Kaminski, T.J. Hubbard, and C. Ortiz. Six weeks of strength and proprioception training does not affect muscle fatigue and static balance in functional ankle instability. *J. Sports Rehabil.* 13:201–227. 2004.
14. Raymakers, J.A., M.M. Samsom, and H.J.J. VerHaar. The assessment of body sway and the choice of the stability parameter(s). *Gait Posture* 21:48–58. 2005.
15. Riemann, B.L. Is there a link between chronic ankle instability and postural instability? *J. Athletic Train.* 37:386–393. 2002.
16. Rozzi, S.L., S.M. Lephart, R. Sterner, and R. Kuligowski. Balance training for persons with functionally unstable ankles. *J. Orthop. Sports Phys. Ther.* 29:478–486. 1999.
17. Sheth, P., B. Yu, E.R. Laskowski, and K. An. Ankle disc training influences reaction times of selected muscles in a simulated ankle sprain. *Am. J. Sports Med.* 25:538–544. 1997.
18. Tropp, H., and P. Odenrick. Postural control in single-limb stance. *J. Orthop. Res.* 6:833–839. 1988.
19. VerHagen, E., A. van der Beek, J. Twisk, L. Bouter, R. Bahr, and W. van Mechelen. The effect of a proprioceptive balance board training program for the prevention of ankle sprains. A prospective controlled trial. *Am. J. Sports Med.* 32:1385–1393. 2004.

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